

Teacher Guide — primary / module-07-eleven-multiplication

IKS Curriculum

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Overview

Module 7 — The Eleven-Friend Trick

Age band: Primary · **Grades:** 3–5 · **Duration:** 10 sessions × 30 min

What children will learn

1. The “Eleven-Friend” pattern: when any 2-digit number meets its “eleven friend,” it grows by adding itself in the middle. ($12 \rightarrow 1_2 \rightarrow 1+2=3 \rightarrow 132$.)
2. The split-and-add trick on no-carry 2-digit numbers (digit-sum < 10).
3. A first peek at the carry case (a “spill-over” — on Day 6, just for awe).
4. A peek at 3-digit × 11 (Day 8) — they watch the teacher; they don’t do it yet.
5. The Sanskrit name *Ekādhikena Pūrveṇa* introduced once, on Day 6, as “the older-kids’ name for what we’re doing.”

How this module sounds

- Lots of chanting. Drumbeat: split, add, drop.
- Story-based — every 2-digit number has an “eleven friend” who makes it bigger.
- No long worksheets. Children call out answers, draw, play.
- Sanskrit term *Ekādhikena Pūrveṇa* introduced once on Day 6.

NEP 2020 alignment

- §5.1 — Foundational numeracy
- §4.6 — Indian Knowledge Systems
- §4.23 — Experiential learning

Materials

- A set of 2-digit number cards 10–98 (for the “meet your eleven friend” game)
- Whiteboards or small slates per child
- Chalk / markers
- A wall poster: “Split. Add. Drop.”

- Counters (optional, for the spill-over demo on Day 6)
- Drawing paper

Day-by-day

Day	Title	What we do
1	Meet Your Eleven-Friend	Story + first try with 12×11 , 23×11
2	Split, Add, Drop chant	Chant + 6 practice numbers
3	The Friend Game (no-carry)	Card-pair game
4	Speed Day	Pair quick-fire
5	What if the middle is too big?	Hint at the carry case
6	Meet the Sanskrit Name	<i>Ekādhikena Pūrveṇa</i> one-line intro
7	Make-a-Game Day	Design a “Make 11” game
8	A Peek at 3-digit $\times 11$	Teacher demo + awe
9	Project Day	Build the game
10	Game Showcase	Play each other’s games

Project

Children design their own “Eleven Friend” card or board game in pairs. They play each other’s on Day 10.

See [assessments/project-brief.md](#).

Sources

The $\times 11$ trick is a mental-math pedagogical technique taught in Indian schools. The Sanskrit name *Ekādhikena Pūrveṇa* comes from Tirthaji’s *Vedic Mathematics* (1965). For older grades’ deeper treatment (including the algebraic proof and the carry case), see the Middle-band version of this module.

10-Day Lesson Plan

Module 7 (Primary) — 10-Day Lesson Plan

At a glance (30-min sessions)

Day	Title	Activity	Output
1	Meet Your Eleven-Friend	Story + first 6 examples	First answers on slate
2	Split, Add, Drop	Chant + practice	Confident chant

Day	Title	Activity	Output
	chant		
3	Friend Game (no-carry)	Card-pair game	First wins
4	Speed Day	Pair quick-fire	Personal best
5	What if the middle is too big?	Hint at carry case	Awe
6	Meet the Sanskrit Name	<i>Ekādhikena Pūrveṇa</i> intro	New word
7	Make-a-Game Day	Design own game	Sketch + first prototype
8	Peek at 3-digit $\times 11$	Teacher demo + watch	Awe
9	Project Day	Build the game	Game ready to play
10	Game Showcase	Play each other's games	Stickers

Daily structure (30 min)

Time	What
4 min	Settle + the “Split, Add, Drop” chant
6 min	Story / new game introduction
15 min	Game / activity
5 min	Sharing / reset

The daily chant

“**Split!** the number. **Add!** the digits. **Drop!** the sum in the middle. That’s the eleven-friend trick!”

Sing it / chorus it with claps. By Day 3 it should be effortless.

Pacing notes

- Games > worksheets at this band. If a worksheet feels slow, drop it and add another game round.
- Day 5 (the carry-case hint) is a “wow” moment — keep it short and mysterious.
- Day 6 names the Sanskrit term ONCE. Don’t drill it.
- Day 8 (peek at 3-digit $\times 11$) is teacher-led demo only. Children watch and gasp; they don’t do it themselves.
- Days 7–10 build to the game showcase. Confirm groups by Day 6.

What children leave with

- Instant recall of any 2-digit $\times 11$ (with digit-sum < 10).
- The “Split, Add, Drop” chant in their bones.
- One Sanskrit word: *Ekādhikena Pūrveṇa* (they don’t need to spell it; they recognise it).
- One game they made and one they played.

- Curiosity about the carry case and 3-digit $\times 11$ for next year.

Teacher Notes

Teacher Notes — Module 7 (Primary)

Posture

Mental $\times 11$ (no-carry) is the single skill we want every child to take home. Everything else — the chant, the game, the Sanskrit name, the peek at 3-digit — is scaffolding around this one win.

Children should leave the module able to: - See **12** and say **132** in one breath. - See **34** and say **374** in one breath. - Recognise that something different happens for **73 $\times$ 11** (without necessarily solving it).

Common things to handle

1. **A child says “What about 73×11 ? You said $7 + 3$ but that’s 10.”** Honour the curiosity. *“Big question — that’s why older kids’ brains get to do more. We’ll see a hint on Day 5.”*
2. **A child can’t yet add single digits fluently.** The trick won’t work until single-digit addition is automatic. Spend 5 extra minutes per class on “make 10” pair recall.
3. **A child knows the trick already.** Make them the “demo kid” — they show the next problem to the class. Refocus their advantage as social, not arithmetic.
4. **A child says “ $1 + 9$ is 10, you can’t write 10 in the middle!”** Brilliant — that’s the carry case. Day 5 covers it; until then, hand-pick examples that avoid it.

Safety

- No physical risk in this module.

Differentiation

- **Tier up:** Grade 5 children may be ready for the carry case ($87 \times 11 = 957$). Pull them aside on Day 5 for the full procedure.
- **Tier down:** Grade 3 children still building single-digit addition need more chant time and slower examples. Skip Day 8’s 3-digit demo if the no-carry case isn’t solid.

Sanskrit pronunciation guide for teachers

Ekādhikena Pūrveṇa = eh-KAA-dhi-keh-na POOR-veh-na.

Say it once on Day 6. Don’t drill it. The win is that children **recognise** the word when they hear it again in Middle band.

Honest framing (for parent questions)

If a parent asks “Is this real Vedic mathematics?” — answer:

“The trick is real and useful. It comes from Tirthaji’s book Vedic Mathematics, published in 1965. Whether the trick is genuinely ancient or a 20th-century synthesis is debated by scholars. Senior band tackles that debate directly. We teach the technique honestly.”

Don’t oversell antiquity. Don’t dismiss it either.

Project oversight

- Days 7–10: children design an “Eleven Friend” game in pairs.
- The game just needs to USE the $\times 11$ trick somewhere (e.g. cards: turn over a 2-digit card; first to call its $\times 11$ wins the card).
- Materials limit: only what’s in the classroom. Don’t expect families to provide game supplies.
- Mixed-age project pairs work well; pair a fluent child with one still building.

If you only have 6 days

- Day 1: Meet Your Eleven-Friend
- Day 2: Chant + Friend Game
- Day 3: Speed Day
- Day 4: Hint at carry case + Sanskrit name
- Day 5: Project Day
- Day 6: Game Showcase

Skip the dedicated 3-digit demo.

Sources

Sources — Module 7 (Primary)

At this band, sources are teacher reference only. No student-facing citations.

Source for the $\times 11$ trick

- **Tirthaji, B. K.** *Vedic Mathematics* (1965), Motilal Banarsidass. Sūtra 1, *Ekādhikena Pūrveṇa*. The book was published posthumously; Tirthaji died in 1960.
- **SutraGanita corpus:** /Users/sree/Projects/SutraGanita/content/ contains the *Microsoft Word - sutras1.pdf* reference (the 16-sūtra list) and *Vedic_Maths.pdf* (ISKCON Desire Tree) with practice examples suitable for primary teachers to pre-read.

Why we don’t teach the full method at Primary

The full $\times 11$ trick requires: - Confident 2-digit-number reading (fine — most Grade 3+ children have this). - Single-digit addition fluency (most have this). - Carry handling (Grade 3 has it; Grade 4–5 is solid).

We teach the no-carry version completely. We *hint* at the carry case on Day 5 (the “spill-over”), and we save the algebraic proof for Middle band.

For families who ask

The $\times 11$ trick is a mental-math technique taught in Indian schools and presented in Tirthaji’s *Vedic Mathematics* (1965). Whether the technique is genuinely ancient (“Vedic”) or a 20th-century synthesis is academically disputed — Senior band (Grade 9–12) tackles this honest framing. At Primary band we just teach the technique enthusiastically.

Day 1 — The Hook — What Is the Trick?

Module: $\times 11$ Multiplication — Ekādhikena Pūrveṇa (Primary) · **Time:** 30 min

Goal

Compute 23×11 , 35×11 mentally using the split-and-sum method.

Materials

- Notebook, pencils, drawing paper
- (See lesson-plan.md for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday’s lesson by asking: “*What’s one thing you remember from yesterday?*”

Today’s idea (10 min)

- *Ekādhikena Pūrveṇa* (IAST: Ekādhikena Pūrveṇa; lit. ““by one more than the previous” — sūtra 1 in Tirthaji”)
- Watch this trick! $23 \times 11 = ?$

Show on the board: “2 _ 3” with a gap. Whisper: “What’s $2 + 3$?” (Wait for “5”.) Fill in: “2 5 3 → 253!”

Each child practices: $12 \times 11 = 132$. $34 \times 11 = 374$. $25 \times 11 = 275$. Try 5 more.

Magic chant: “*Split, add, put in middle!*” (Children chant 3 times.)

Activity (8 min)

A simplified version of *Mental Math Sprint Cards* (see [activities/](#)). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See homework.md.)

Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Day 2 — The Carry Case

Module: $\times 11$ Multiplication — Ekādhikena Pūrveṇa (Primary) · **Time:** 30 min

Goal

Compute 87×11 , 65×11 correctly using the carry rule.

Materials

- Notebook, pencils, drawing paper
- (See lesson-plan.md for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: “*What's one thing you remember from yesterday?*”

Today's idea (10 min)

- *sūtra* (IAST: sūtra; lit. “aphorism / formula”)
- What if the middle sum is too big? Like 87×11 .

Show: “8 _ 7” with gap. $8 + 7 = 15$. Whisper: “15 is two digits! Keep the 5, carry the 1.”

Result: $8 + 1 = 9$, then 5, then 7 → **957**.

Practice: $67 \times 11 = 7\ 3\ 7$. $95 \times 11 = 1\ 0\ 4\ 5$. “*Carry the tens, keep the ones!*” (Chant.)

Activity (8 min)

A simplified version of *Mental Math Sprint Cards* (see [activities/](#)). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See [homework.md](#).)

Teacher notes

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- If energy is low, swap the activity for free drawing.

Day 3 — Why It Works — The Algebra

Module: $\times 11$ Multiplication — Ekādhikena Pūrveṇa (Primary) · **Time:** 30 min

Goal

Derive $(10a+b)\times 11 = 100a + 10(a+b) + b$ and explain.

Materials

- Notebook, pencils, drawing paper
- (See [lesson-plan.md](#) for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: “*What's one thing you remember from yesterday?*”

Today's idea (10 min)

- *Anurūpyena* (IAST: Anurūpyena; lit. “proportionately” — sūtra used for $\times 12$, $\times 13$ extensions”)
- Why does the magic work?

Show 23×11 with paper strips: - One strip of 23 + ten strips of 23 = 11 strips. - Stack them: 230 + 23 = 253. - The “230” gives the 2 and the 3 (separated). The “23” gives the +2 in the tens and +3 in the ones. - $2 + 3 = 5$ in the tens place. The 2 and 3 stay on the ends!

Magic? No — just adding! Children try with strips of 14: $14 \times 11 = 140 + 14 = 154$.

Activity (8 min)

A simplified version of *Mental Math Sprint Cards* (see [activities/](#)). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See [homework.md](#).)

Teacher notes

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- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Day 4 — 3-Digit $\times$ 11

Module: $\times 11$ Multiplication — Ekādhikena Pūrveṇa (Primary) · **Time:** 30 min

Goal

Compute $234 \times 11 = 2_5_7_4$ using the extended split method.

Materials

- Notebook, pencils, drawing paper
- (See [lesson-plan.md](#) for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: “*What's one thing you remember from yesterday?*”

Today's idea (10 min)

- *Ekādhikena Pūrveṇa* (IAST: Ekādhikena Pūrveṇa; lit. ““by one more than the previous” — sūtra 1 in Tīrthajī”)
- For 3-digit numbers: split with TWO gaps!

$234 \times 11 = ?$ - Write “2 _ _ 4” with two gaps. - Fill: $2+3 = 5$ (first gap), $3+4 = 7$ (second gap). - Answer: **2574**.

Practice: $123 \times 11 = 1 (1+2) (2+3) 3 = 1353$. $456 \times 11 = 5016$ (with carries — hard).

Most children stick to no-carry cases this day.

Activity (8 min)

A simplified version of *Mental Math Sprint Cards* (see [activities/](#)). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See [homework.md](#).)

Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Day 5 — Speed Day

Module: $\times 11$ Multiplication — Ekādhikena Pūrveṇa (Primary) · **Time:** 30 min

Goal

Complete a 50-problem mental-math sprint in 5 minutes.

Materials

- Notebook, pencils, drawing paper
- (See [lesson-plan.md](#) for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday’s lesson by asking: “*What’s one thing you remember from yesterday?*”

Today’s idea (10 min)

- *sūtra* (IAST: sūtra; lit. “aphorism / formula”)

- Speed time! Today we play a quick $\times 11$ game.

In pairs: one child says a 2-digit number, the other shouts the answer $\times 11$. Switch every 30 seconds. 5 rounds.

End with class cheer: “*Split! Add! Middle! Done!*”

Activity (8 min)

A simplified version of *Mental Math Sprint Cards* (see [activities/](#)). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See [homework.md](#).)

Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Day 6 — Anurūpyena — $\times 12$ to $\times 19$

Module: $\times 11$ Multiplication — Ekādhikena Pūrveṇa (Primary) · **Time:** 30 min

Goal

Extend the method to $\times 12$ and $\times 13$ using one-more-digit adjustments.

Materials

- Notebook, pencils, drawing paper
- (See [lesson-plan.md](#) for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday’s lesson by asking: “*What’s one thing you remember from yesterday?*”

Today's idea (10 min)

- *Anurūpyena* (IAST: Anurūpyena; lit. “proportionately” — sūtra used for $\times 12$, $\times 13$ extensions)
- We can do $\times 12$ too! Just split into $\times 10 + \times 2$.
- $15 \times 12 = (15 \times 10) + (15 \times 2) = 150 + 30 = \mathbf{180}$.

Practice in pairs: one says a number, the other doubles it for $\times 2$, then adds a zero for $\times 10$, then adds the two together for $\times 12$.

Try $\times 13$, $\times 14$ only if time. Most children stick to $\times 12$ today.

Activity (8 min)

A simplified version of *Mental Math Sprint Cards* (see [activities/](#)). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See [homework.md](#).)

Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Day 7 — Limits of the Method

Module: $\times 11$ Multiplication — Ekādhikena Pūrveṇa (Primary) · **Time:** 30 min

Goal

Recognise when traditional long multiplication is faster than the sūtra method.

Materials

- Notebook, pencils, drawing paper
- (See [lesson-plan.md](#) for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: “*What’s one thing you remember from yesterday?*”

Today’s idea (10 min)

- *Ekādhikena Pūrveṇa* (IAST: Ekādhikena Pūrveṇa; lit. ““by one more than the previous” — sūtra 1 in Tirthaji”)
- The trick is special! It works best for $\times 11$ (and $\times 12$).
- For things like $\times 23$ or $\times 47$, the trick is hard. Then we use the regular way.

Show: 23×47 . “*Would the trick work?*” (No — 47 is too far from 10.) “*So we do it the long way.*”

Children practice both methods on 24×11 vs 24×23 to feel the difference.

Activity (8 min)

A simplified version of *Mental Math Sprint Cards* (see [activities/](#)). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See [homework.md](#).)

Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Day 8 — Tirthaji and Historical Context

Module: $\times 11$ Multiplication — Ekādhikena Pūrveṇa (Primary) · **Time:** 30 min

Goal

Explain who Tirthaji was and why scholars debate the “Vedic” attribution.

Materials

- Notebook, pencils, drawing paper
- (See [lesson-plan.md](#) for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: “*What's one thing you remember from yesterday?*”

Today's idea (10 min)

- *sūtra* (IAST: sūtra; lit. “aphorism / formula”)
- The trick comes from a book by Bharati Krishna Tirthaji, who lived about 100 years ago.
- He named it after old Sanskrit ideas. Today, scholars are not sure if the methods are really thousands of years old. But the *trick still works* — and that's what matters for us!

Show a photo (or drawing) of Tirthaji. Children put a thumbs-up if they like the trick.

Activity (8 min)

A simplified version of *Mental Math Sprint Cards* (see [activities/](#)). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See [homework.md](#).)

Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Day 9 — Mixed Practice

Module: ×11 Multiplication — Ekādhikena Pūrveṇa (Primary) · **Time:** 30 min

Goal

Solve a mixed problem set using best-fit method per problem.

Materials

- Notebook, pencils, drawing paper
- (See [lesson-plan.md](#) for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: “*What's one thing you remember from yesterday?*”

Today's idea (10 min)

- *Anurūpyena* (IAST: Anurūpyena; lit. “proportionately” — sūtra used for $\times 12$, $\times 13$ extensions”)
- Today we play a sorting game!

Show 10 problems on cards. For each, children hold up a sign: - “Trick!” (for $\times 11$ problems) - “Long!” (for harder problems)

Then solve 5 $\times 11$ problems for practice. End with applause.

Activity (8 min)

A simplified version of *Mental Math Sprint Cards* (see [activities/](#)). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See [homework.md](#).)

Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Day 10 — Final Assessment

Module: $\times 11$ Multiplication — Ekādhikena Pūrveṇa (Primary) · **Time:** 30 min

Goal

Take the summative quiz and reflect on which method works best.

Materials

- Notebook, pencils, drawing paper

- (See lesson-plan.md for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: “What’s one thing you remember from yesterday?”

Today’s idea (10 min)

- *Ekādhikena Pūrveṇa* (IAST: Ekādhikena Pūrveṇa; lit. ““by one more than the previous” — sūtra 1 in Tirthajī”)
- Final day! We celebrate the $\times 11$ trick.

Each child shows their best $\times 11$ problem to a partner. Cheer for each one.

Take the small quiz with stickers (every child gets a “Number Magician” sticker).

End: class chant — “*Split! Add! Middle! Magic!*”

Activity (8 min)

A simplified version of *Mental Math Sprint Cards* (see activities/). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See homework.md.)

Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Formative Quiz — $\times 11$ Multiplication — Ekādhikena Pūrveṇa (Primary)

When: mid-module (after Day 5). **Duration:** 20 minutes. **Format:** mix of short answer, multiple choice, and reflection. Sticker self-assessment for Primary.

Questions

1. Name the three / four key terms introduced in this module so far. Give the IAST and one-line meaning of each.
2. State the central idea of $\times 11$ Multiplication — *Ekādhikena Pūrveṇa* in one sentence.
3. (Multiple choice) Which day in this module focused on ‘Why It Works — The Algebra’?
(a) Day 1 (b) Day 2 (c) Day 3 (d) Day 4
4. True / False — and explain in one sentence: *$\times 11$ Multiplication — Ekādhikena Pūrveṇa is purely a religious concept and has no scientific framing.*
5. Connect $\times 11$ Multiplication — *Ekādhikena Pūrveṇa* to **one** other module in this curriculum. Which module, and how?
6. List **two** sources cited so far in this module.
7. (Short answer) Why does the module distinguish ‘*Ekādhikena Pūrveṇa*’ from related concepts? Give one example.
8. (Short answer) In your own words, explain Day 3’s concept.
9. Identify **one** common misconception about $\times 11$ Multiplication — *Ekādhikena Pūrveṇa* and rebut it in one sentence.
10. Write **one** question you have about this module so far — something you want clarified before the summative.

How this is used

This quiz is **formative** — it’s a check-in to see what’s landing, not a grade. Teacher reviews answers to adjust pacing for Days 6–10. Students may swap and self-mark.

Summative Quiz — $\times 11$ Multiplication — *Ekādhikena Pūrveṇa* (Primary)

When: end of module (after Day 10). **Duration:** 30 minutes. **Total points:** 20.

Questions

Part A — Vocabulary (~5 points) A1. Define *Ekādhikena Pūrveṇa* in your own words. A2. Define *sūtra* in your own words. A3. Define *Anurūpyena* in your own words.

Part B — Concepts (~10 points) B1. Explain the central idea of $\times 11$ Multiplication — *Ekādhikena Pūrveṇa* in your own words (3–4 sentences). B2. Describe the method or framework students used in Day 4 of this module. B3. Identify ONE source cited in this module and explain why it matters. B4. Compare $\times 11$ Multiplication — *Ekādhikena Pūrveṇa* with a concept from another module (your choice). What’s similar, what’s different? B5. Identify ONE common misconception about this module and rebut it in 2–3 sentences.

Part C — Application (~5 points) C1. Give an example of $\times 11$ Multiplication — *Ekādhikena Pūrveṇa* from your own daily life. C2. Describe one way this module connects to a current real-world issue (or scientific question).

Part D — Reflection (~5 points; ungraded for content) D1. What surprised you most in this module? D2. What’s one question you still have?

Marking guidance

- Vocabulary: give credit for correct meaning, even if exact wording differs.
- Concepts: look for evidence of understanding, not memorisation.
- Application: any reasonable example accepted.
- Reflection: ungraded for content; awarded for thoughtfulness.

Pass mark

A score of 12/20 indicates the student has met the module’s learning objectives.

Homework

Homework — $\times 11$ Multiplication — Ekādhikena Pūrveṇa (Primary)

Light daily tasks. Each takes 5–10 minutes.

1. **Day 1 — The Hook — What Is the Trick?** · Draw about today’s concept. (~5 min)
2. **Day 2 — The Carry Case** · Draw about today’s concept. (~5 min)
3. **Day 3 — Why It Works — The Algebra** · Draw about today’s concept. (~5 min)
4. **Day 4 — 3-Digit $\times 11$** · Draw about today’s concept. (~5 min)
5. **Day 5 — Speed Day** · Draw about today’s concept. (~5 min)
6. **Day 6 — Anurūpyena — $\times 12$ to $\times 19$** · Draw about today’s concept. (~5 min)
7. **Day 7 — Limits of the Method** · Draw about today’s concept. (~5 min)
8. **Day 8 — Tirthaji and Historical Context** · Draw about today’s concept. (~5 min)
9. **Day 9 — Mixed Practice** · Draw about today’s concept. (~5 min)
10. **Day 10 — Final Assessment** · Draw about today’s concept. (~5 min)

Notes

- Homework is intended to consolidate what was learned in class, not introduce new content.
- If a child struggles, the parent should consult the teacher rather than push.
- Capstone project days (Module 15) may have lighter daily homework so students can focus on the project.

Activity 1 — Mental Math Sprint Cards

Module: $\times 11$ Multiplication — Ekādhikena Pūrveṇa · **Band:** Primary · **Duration:** 20 min

Purpose

Reinforce the day-by-day concepts of $\times 11$ Multiplication — Ekādhikena Pūrveṇa through a hands-on activity. Aligned with NEP 2020 §4.23 (experiential learning).

Materials

flashcards (20–50 pre-printed $\times 11$ problems), stopwatch

What students do

In pairs: one student holds up a card, the other answers within 3 seconds using the $\times 11$ method. Swap. Track personal best time.

Step-by-step

1. **Setup (5 min)** — gather materials; explain the task.
2. **Working phase (10 min)** — students work individually or in pairs.
3. **Share (5 min)** — selected students share their results with the class.

Differentiation

- **Extension:** ask advanced students to do an additional iteration or analysis.
- **Support:** pair students who need scaffolds; offer partial templates.

Assessment

This activity is **formative**. The teacher observes participation, accuracy, and reflection quality. No grade is assigned.

Safety

- General safety briefing before any materials are handled.
- If herbs, plants, or food are involved: kitchen-amount only, no ingestion of unknown preparations.
- Outdoor activities: stay in the designated area; supervised by the teacher.

Project Brief

Project Brief — Method Comparison Poster

Module: $\times 11$ Multiplication — Ekādhikena Pūrveṇa · **Band:** Primary

What you will do

Choose 5 multiplication problems and solve them using both the $\times 11$ method and traditional long multiplication. Time yourself. Make a poster showing when each method is faster and why.

Why

This project asks you to apply concepts from this module to your own life or a real situation. It is graded on the rubric in `rubric.md`.

Deliverables

- **One-page project plan** (Day 2 of the module): problem, sources, deliverable, timeline.
- **Daily journal entries** (one per day, starting Day 6): what you did, what worked, what didn't.
- **Final artefact** (a drawing / made object).
- **Written reflection** (oral share): what you learned, what surprised you, what you're still uncertain about.
- **Presentation** (3-minute share with the class).

Timeline

- **Day 1–2:** scope and plan.
- **Day 3–4:** research.
- **Day 5:** mid-project critique.
- **Day 6–7:** build / make.
- **Day 8:** document.
- **Day 9:** rehearse.
- **Day 10:** share day.

Sources

You must cite at least 1 sources. Use the form: *Author, Title, year*. (See `../sources.md` for examples.)

Safety

- Any project involving plants / food / outdoor work must be supervised.
- **No herbal medicine recommendations.** Modules 10 and 4 give awareness; therapeutic decisions belong to qualified practitioners.

Grading

See `rubric.md`. The project is worth the same weight as the summative quiz.

Rubric

Assessment Rubric — ×11 Multiplication — Ekādhikena Pūrveṇa (Primary)

This rubric applies to the project (see `project-brief.md`) and may also inform the summative quiz.

Criteria

- **Knows the new word(s)** — 🌟 Strong understanding / 🌱 Growing / ☁️ Needs more practice
- **Joined in activities** — 🌟 Strong understanding / 🌱 Growing / ☁️ Needs more practice
- **Asked good questions** — 🌟 Strong understanding / 🌱 Growing / ☁️ Needs more practice
- **Drew / made something** — 🌟 Strong understanding / 🌱 Growing / ☁️ Needs more practice

How the teacher uses this rubric

1. After the project share, score each student on each criterion.
2. Convert to a band (4 = Exceeding, 3 = Meeting, etc.) for the report card.
3. Share at least one strength and one area for growth with each student.

Self-assessment (Senior students)

Senior students complete a parallel self-assessment using the same rubric before the teacher's assessment. The teacher and student then meet to compare.